

Autumn (Odd) Semester Examination -2025

Course Code: PHYUGMCC2305

Course Title: Thermal Physics

Department of Appearing Students: PHYSICS, SEM-III, 2nd Year

Full Marks: 60

Time: 2:30 hrs.

[Use separate sheets for answering Group-A and Group-B questions.]

GROUP- A

(Answer any FIVE questions)

1. (a) The velocity distribution function of a group of N particles is given by

$$dN_v = kv^2 dv \quad \text{for } 0 \leq v \leq V$$

$$= 0 \quad \text{for } v > V$$

Calculate the constant k in terms of N and V , and also calculate the mean velocity and RMS velocity of the particle. 2+2+2

2. (a) Write down the van der Waals equation of state for a real gas and explain the significance of the correction terms. (b) Find the critical constants (V_c , P_c , T_c) of a real gas obeying the van der Waals equation of state? (1+2) +3
3. (a) Show that with the increase of volume per mole, the van der Waals equation of state tends to that of an ideal gas. (b) What is the Boyle temperature? How is it related to critical temperature? 2+(1+3)
4. (a) A rigid triangular molecule consists of three non-collinear atoms joined by rigid rods. Using the laws of equipartition of energy, obtain the average energy and the constant-pressure molar specific heat C_p of an ideal gas consisting of such molecules.
- (b) In 1- dimension, an ensemble of N classical particles has energy of the form, $E = \frac{1}{2m}(p_x^2) + \frac{1}{2}kx^2$ where m is the mass of the particle, k is a constant. Find the average internal energy of the system at temperature T . 3+3
5. (a) Write down the expression (no derivation) for the coefficient of viscosity and thermal conductivity of a gas and find the relation between them. (b) Explain how the coefficient of viscosity (η) varies with Pressure and temperature? Explain. 2+ (2+2)
6. Two boxes, A and B, contain an equal number of molecules of the same gas. If the volumes are V_A and V_B , find the ratio of their respective mean free paths. (b) A narrow molecular beam enters a vessel filled with gas under low Pressure. Find the mean free path of the molecules if the beam intensity decreased to $1/n$ times its initial intensity over a distance Δl . 3+3

3x3=2

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GROUP- B

(Answer Question No. 7 and any three from the remaining)

7. a) An oil bath kept at 27°C is being supplied heat at the rate of 100Js^{-1} . Assuming the process to be quasi-static, calculate the rate of increase of entropy of the system. 4
- b) With the use of a suitable diagram prove that working between same two temperatures no other engine can be more efficient than a reversible Carnot engine. 3
- c) An engine absorbs heat at temperature of 1000K and rejects heat at 600K . Assuming the engine operating at maximum possible efficiency, estimate the amount of work performed by the engine for 200J heat input. 3
- d) Check whether the relations $\left(\frac{\partial H}{\partial S}\right)_P = T$ and $\left(\frac{\partial H}{\partial P}\right)_S = V$ hold true or not for the thermodynamical potential called enthalpy, $H = U + PV$. 2
- e) From the first law of thermodynamics show that for an ideal gas $C_P - C_V = R$, where symbols have their usual meanings. 3 } 5
8. With a suitable diagram of Carnot cycle, derive an expression for the efficiency of car reversible Carnot engine explaining all the processes involved in the cycle. Can the efficiency of a Carnot engine be 100%? Justify your response. 4+1 = 5
9. When a mass m of water at temperature $T_1\text{K}$ is isobarically and adiabatically mixed with an equal amount of water at $T_2\text{K}$, then show that the entropy change of the universe is $2mC_p \log \left(\frac{T_1+T_2}{2\sqrt{T_1T_2}}\right)$ ✓ 5
10. With a suitable diagram explain the thermodynamic processes involved in the diesel cycle. How a diesel cycle differs from an Otto cycle? 4+1
11. Derive Maxwell's thermodynamical relations: $\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$ and $\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$. You may use the expressions of thermodynamical potentials: $F=U-TS$ and $G=H-TS$ respectively. Symbols have their usual meanings. 2+3
12. a) What are extensive in intensive variables? (b) What do you mean by a thermodynamic scale of temperature? Is it possible to attain absolute zero on the thermodynamic scale of temperature? Justify your answer. 2+(2+1)